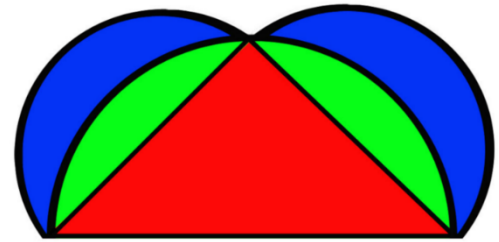


Senior Mathematics Competition 2025

Thursday 31st July 2025

Time Allowed: 90 minutes

Total: 42 marks



NZAMT
Te Rōpū Kaiako Pāngarau o Aotearoa
New Zealand Association of Mathematics Teachers
est. 1978

Instructions

- Attempt as many questions as possible. It is unlikely you will finish them all.
- Full working should accompany all solutions on paper.
- Label each question clearly and put your name(s) on each page.
- Approved NCEA calculators may be used, but no other reference material is permitted.
- Devices/phones are not permitted during the competition.

Question One

(2 marks)

$$7! = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1.$$

$$7!! = 7 \times 5 \times 3 \times 1.$$

Calculate: $9! + 8!! + 7!!!$

Question Two

(2 marks)

$$L^2 + O^2 = 122$$

$$L + O = 12$$

$$V^2 + E^2 = 170$$

$$V + E = 14$$

What is $L.O.V.E$?

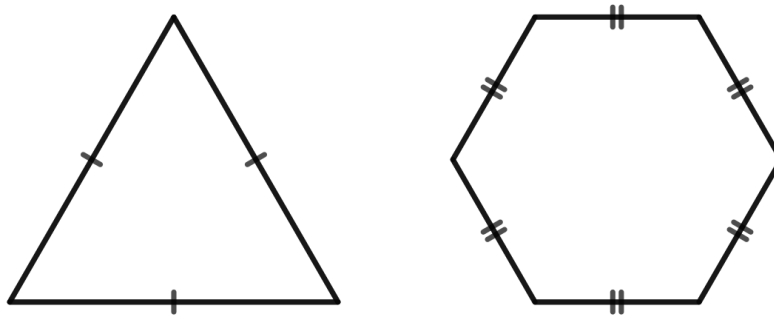


Question Three**(2 marks)**

$$\frac{n}{9!} + \frac{1}{8!} = \frac{100}{10!}$$

Calculate the value of n .**Question Four****(3 marks)**

Suppose an equilateral triangle and a regular hexagon have the same perimeter. Find the simplest ratio of the areas of their respective inscribed circles.

**Question Five****(2 marks)**

Expand and simplify $(\sqrt{2} - 1)^6$, leaving your answer in the form of $a \pm b\sqrt{c}$.

Question Six**(2 marks)**

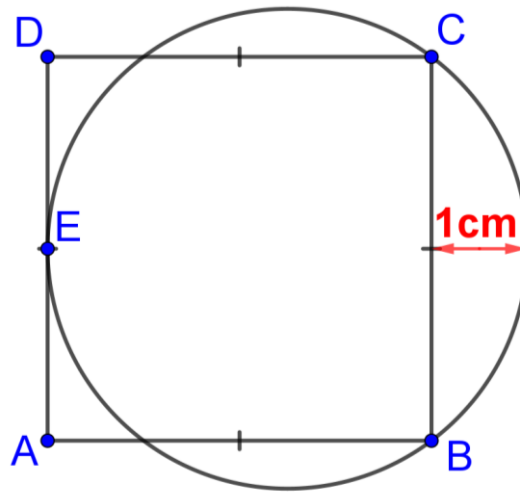
A two-headed coin (heads on both sides), a two-tailed coin (tails on both sides), and an ordinary coin (heads and tails) are placed in a bag. The bag is shaken and one of the coins is drawn at random and flipped; it comes up “heads.” What is the probability that there is a head on the other side of this coin? Explain carefully.

**Question Seven****(2 marks)**

Find the smallest positive numbers a and b for which $7a^3 = 11b^5$.

Question Eight**(3 marks)**

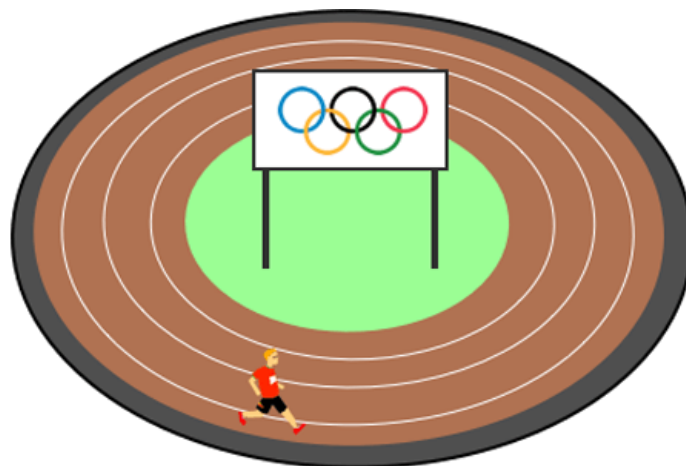
Suppose points B, C and E are common points to the square ABCD and the circle shown below. E is a point of tangency. Find the exact area of the square ABCD.

**Question Nine****(5 marks)**

Solve the following equation: $(x^2 - 6x + 10)^{(x^2 - 8x + 16)} = 1$, by identifying three different scenarios and finding any real solutions to each of these scenarios.

Question Ten**(3 marks)**

Two runners start together on a circular track, running at different constant speeds. If they head in opposite directions they meet after a minute; if they head in the same direction, they meet after an hour. What is the ratio of their speeds?

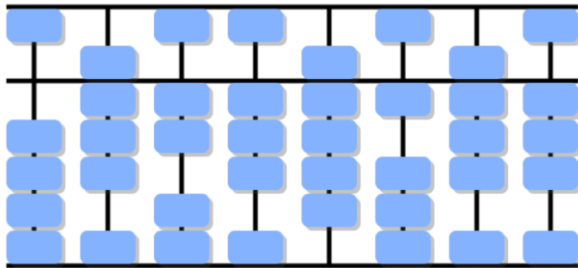


Question Eleven**(3 marks)**

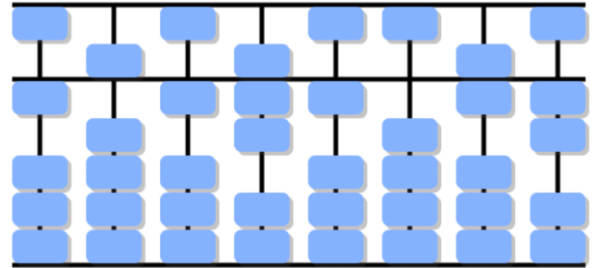
Does $4^9 + 6^{10} + 3^{20}$ happen to be a prime number? Explain carefully.

Question Twelve**(1 mark)**

Using the following abacus, calculate the sum of:



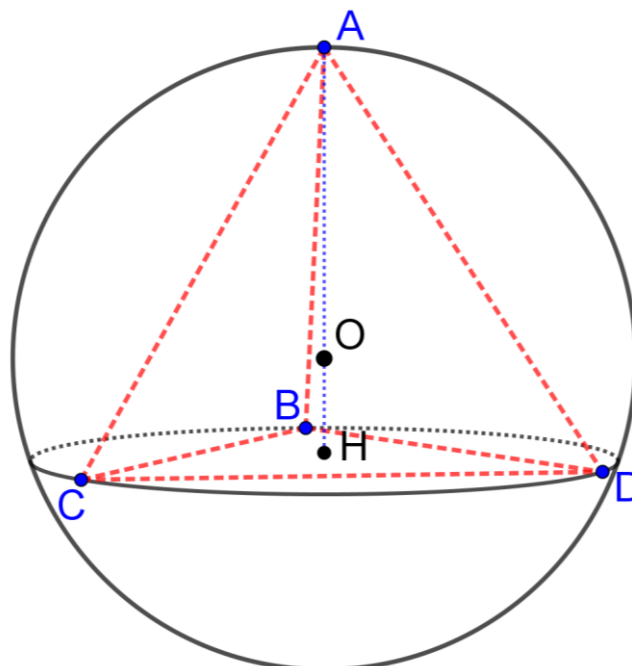
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**Question Thirteen****(3 marks)**

Solve the equation: $x! = x^3 - x$, given $x \in W$

Question Fourteen**(4 marks)**

O is the centre of a sphere. A regular tetrahedron with vertices A, B, C and D is inscribed within the circumsphere. A circumsphere is a sphere with a polyhedron inscribed inside and all vertices touching the surface of the sphere. Point H is the centroid of the base of the tetrahedron. Find the length of the radius OA for the circumsphere of a regular tetrahedron with an edge length of a . Note that the ratio of OA : OH is 3 : 1.



Question Fifteen**(3 marks)**

Solve the following equation: $\sqrt{\sqrt{x} + 15} + \sqrt{\sqrt{x}} = 15$

Question Sixteen**(2 marks)**

Insert the numbers 2 to 7 into the remaining 6 circles, so that for any particular circle, the sum of the numbers in each connecting circle sum to the value indicated in the table.

Number	Sum
1	10
2	20
3	15
4	13
5	21
6	3
7	8
8	14

